Close

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1.Perform the following operations using Python on suitable data sets:

Read data from different formats (like csv, xls) Find Shape of Data Find Missing Values Find data type of each column Finding out Zero's Indexing and selecting data, sort data, Describe attributes of data, checking data types of each column, counting unique values of data, format of each column, converting variable data type (e.g. from long to short, vice versa)

```
import pandas as pd

data = {
    'ID': [1, 2, 3, 4, 5],
    'Name': ['soha', 'kanti', 'Jmaui', 'Nsenn', 'Arun'],
    'Marks': [85, 0, 78, None, 92],
    'Grade': ['A', 'F', 'B', 'B', 'A']
}

df = pd.DataFrame(data)
df
```

	ID	Name	Marks	Grade
0	1	soha	85.0	A
1	2	kanti	0.0	F
2	3	Jmaui	78.0	B
3	4	Nsenn	NaN	B
4	5	Arun	92.0	A

Next steps: Generate code with df New interactive sheet

Read Data from Different Formats (CSV & Excel)

```
df.to_csv('student.csv', index=False)
df.to_excel('student.xlsx', index=False)

df_csv = pd.read_csv('student.csv')
df_xls = pd.read_excel('student.xlsx')
```

Find Shape of Data

```
df.shape
```

```
(5, 4)
```

Find Missing Values

```
df.isnull().sum()
```

	0
ID	0
Name	0
Marks	1
Grade	0

dtype: int64

Find Data Type of Each Column

```
df.dtypes
```

	0
ID	int64
Name	object
Marks	float64
Grade	object

dtype: object

Finding zero's

```
(df['Marks'] == 0).sum()
```

```
np.int64(1)
```

```
(df == 0).sum()
```

	0
ID	0
Name	0
Marks	1
Grade	0

dtype: int64

Indexing and Selecting data

```
df['Name']
```

	Name
0	soha
1	kanti
2	Jmaui
3	Nsenn
4	Arun

dtype: object

```
df.loc[2]
```

	2
ID	3
Name	Jmaui
Marks	78.0
Grade	B

dtype: object

```
df.loc[1, 'Marks']
```

```
np.float64(0.0)
```

Sorting data

```
df.sort_values(by='Marks')
```

	ID	Name	Marks	Grade
1	2	kanti	0.0	F
2	3	Jmaui	78.0	B
0	1	soha	85.0	A
4	5	Arun	92.0	A
3	4	Nsenn	NaN	B

```
df.sort_values(by='Marks', ascending=False)
```

	ID	Name	Marks	Grade
4	5	Arun	92.0	A
0	1	soha	85.0	A
2	3	Jmaui	78.0	B
1	2	kanti	0.0	F
3	4	Nsenn	NaN	B

Describe Attributes

```
df.describe()
```

	ID	Marks
count	5.000000	4.000000
mean	3.000000	63.750000
std	1.581139	42.882592
min	1.000000	0.000000
25%	2.000000	58.500000
50%	3.000000	81.500000
75%	4.000000	86.750000
max	5.000000	92.000000

Check Datatype

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 5 entries, 0 to 4
Data columns (total 4 columns):
 #   Column  Non-Null Count  Dtype
---  ------  --------------  -----
 0   ID      5 non-null      int64
 1   Name    5 non-null      object
 2   Marks   4 non-null      float64
 3   Grade   5 non-null      object
dtypes: float64(1), int64(1), object(2)
memory usage: 292.0+ bytes
```

Counting Unique Values

```
df['Grade'].value_counts()
```

Grade	count
A	2
B	2
F	1

dtype: int64

```
df['Name'].nunique()
```

```
5
```

Format Each Column

```
df.dtypes
```

	0
ID	int64
Name	object
Marks	float64
Grade	object

dtype: object

Converting datatype

```
df['Marks'] = df['Marks'].fillna(0).astype(int)
```

```
df['ID'] = df['ID'].astype(str)
```

```
df.dtypes
```

	0
ID	object
Name	object
Marks	int64
Grade	object

dtype: object

```
df
```

	ID	Name	Marks	Grade
0	1	soha	85	A
1	2	kanti	0	F
2	3	Jmaui	78	B
3	4	Nsenn	0	B
4	5	Arun	92	A

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